

A 12-to-20Gb/s, 2.93-pJ/bit Jitter-Filtering Retimer with High Input Jitter Tolerance in 28nm CMOS

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Abstract—This paper presents a high-speed jitter-filtering retimer operating from 12 to 20 Gb/s in 28nm CMOS technology. The design addresses the trade-off between input jitter tolerance (JTOL) and output jitter suppression by cascading a wide-bandwidth clock and data recovery (CDR) circuit with a narrow-bandwidth phase-locked loop (PLL). An injection-locked oscillator (ILO)-based CDR enables high input bandwidth and superior tracking of incoming jitter, while a second-order PLL filters high-frequency jitter components. The system achieves a JTOL bandwidth exceeding 500 MHz and demonstrates 1 UIPP JTOL at 173 MHz at 20 Gb/s, with an energy efficiency of 2.93 pJ/bit.

Index Terms—Jitter filtering, retimer, clock and data recovery (CDR), phase-locked loop (PLL), injection-locked oscillator (ILO), high-speed serial links

I. INTRODUCTION

With the increasing demand for high-speed serial communication in applications such as automotive Ethernet and high-performance computing, signal integrity degradation due to inter-symbol interference (ISI) and jitter has become a critical challenge. Retimers are widely used to recover and regenerate degraded signals across long interconnects.

Conventional CDR architectures suffer from a fundamental trade-off: wide bandwidth improves jitter tolerance but allows more jitter to propagate to the output, while narrow bandwidth filters jitter but reduces tracking capability. This work addresses this limitation by combining a wide-bandwidth CDR with a narrow-band PLL, enabling both high JTOL and effective jitter suppression.

II. MODULATOR DESIGN AND IMPLEMENTATION

The proposed retimer architecture consists of three main blocks: a wide-bandwidth input CDR, a jitter-filtering PLL, and a transmitter (TX) stage.

The input CDR employs an injection-locked oscillator (ILO) to achieve rapid phase alignment without requiring a loop filter, thereby enabling wide input bandwidth. A delay line and pulse generator control the injection timing, ensuring proper alignment between the data and clock edges. The delay-locked loop (DLL) adjusts the delay such that the total delay converges to one unit interval (UI).

Frequency locking is achieved by comparing the ILO oscillation period with the input data rate, allowing the oscillator to converge to the desired frequency.

The jitter-filtering PLL utilizes an LC-based voltage-controlled oscillator (LC-VCO) to achieve low phase noise and narrow loop bandwidth. This enables effective filtering of high-frequency jitter components tracked by the CDR. The PLL includes charge pump and loop filter components for bandwidth optimization.

The transmitter employs a quarter-rate serializer with an unstacked multiplexer architecture, minimizing bandwidth degradation. A tail-less output driver improves output bandwidth and efficiency while supporting high-speed operation.

III. EXPERIMENTAL RESULTS AND SYSTEM DEMONSTRATION

The retimer is implemented in a 28nm CMOS process with an active area of approximately 0.4 mm². The measurement setup utilizes a bit error rate tester (BERT) to inject sinusoidal jitter into the input signal and evaluate system performance.

Measurement results demonstrate:

- Operation up to 20 Gb/s data rate
- JTOL of 1 UIPP at 173 MHz (20 Gb/s)
- JTOL bandwidth exceeding 500 MHz (limited by test equipment)
- 0.57 UIPP JTOL at 500 MHz

Eye diagram measurements show a significantly improved output eye compared to a closed input eye under high jitter conditions, confirming effective jitter filtering. The PLL output clock exhibits reduced phase noise compared to the recovered CDR clock.

The system achieves an overall energy efficiency of 2.93 pJ/bit, demonstrating a favorable trade-off between performance and power consumption.

IV. CONCLUSION

This work presents a jitter-filtering retimer that overcomes the traditional trade-off between jitter tolerance and jitter suppression. By cascading an ILO-based wide-bandwidth CDR with a narrow-band PLL, the design achieves both high JTOL and effective jitter filtering.

The prototype demonstrates 20 Gb/s operation with JTOL bandwidth exceeding 500 MHz and 1 UIPP tolerance at 173 MHz, along with competitive energy efficiency. This architecture is well-suited for next-generation high-speed serial communication systems.